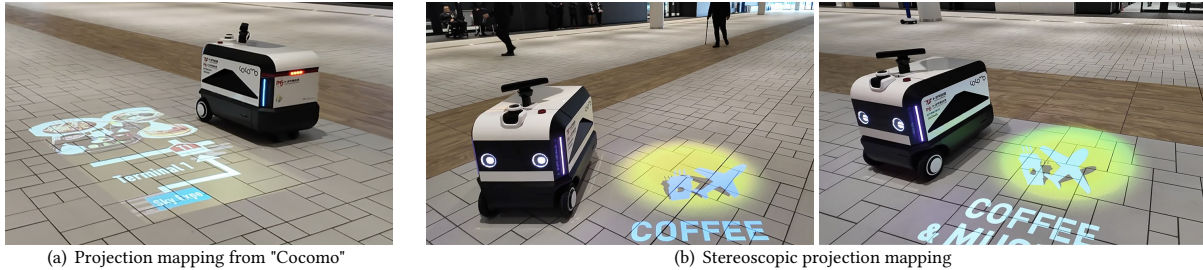


Development of a stereoscopic projection mapping system using a small mobile robot

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(a) Projection mapping from "Cocomo"

(b) Stereoscopic projection mapping

Figure 1: A stereoscopic projection mapping system using a small mobile robot "Cocomo".

KEYWORDS

projection mapping, stereoscopic image, anamorphosis

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1 INTRODUCTION

In this study, we have developed a stereoscopic projection mapping system using a small mobile robot with the aim of providing information and promotion to people in public spaces and events (Figure 1). This system incorporates a projector inside a small mobile robot capable of autonomous movement, and projects images onto the floor while moving (Figure 1(a)). When the system detects a person in front of the robot, it stops moving and projects stereoscopic images for the person while facing them (Figure 1(b)).

The mobile small robot can recognize its own position, so the projected images automatically change according to their position. Additionally, it collects information about the crowdedness of surrounding restaurants via the system that visualize visitor behavior within facilities which has been released as a service by Toyota Boshoku Corporation, and project advertisements for restaurants with available space.

VJ Suave developed a projection mapping system with a projector mounted on a small vehicle[VJSUAVE 2019]. However our system

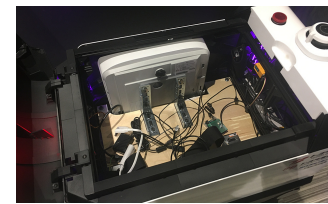


Figure 2: Installing a projector inside Cocomo.

has a unique feature not found in conventional systems, which is projecting stereoscopic images while tracking a person.

2 METHODS

The mobile small robots used are called "Cocomo", developed by Toyota Motor East Japan, Inc. It has dimensions of length 88(cm), width 57(cm), and height 65(cm), with a box-shaped body equipped with four wheels. In addition to GPS, Cocomo is equipped with a 2D LiDAR, allowing it to estimate its own position through SLAM.

We installed a short-throw WUXGA (1920×1200) projector and a PC (Mac mini) inside Cocomo (Figure 2). This setup enables projecting images onto the floor with dimensions of 194×121(cm). The projector has a brightness of 4500(lm), ensuring sufficient brightness for projecting clear images even in illuminated indoor environments. In addition, we have also installed a Kinect camera to obtain the 3D positions of people in front of Cocomo.

Cocomo moves automatically along a pre-set route while estimating its own position with SLAM. At that time, it projects images onto the side floor using installed projectors. We projected videos introducing event venues, restaurants, and shops, considering conducting a demonstration experiment at the exhibition hall. The displayed content varied depending on Cocomo's position. For example, when Cocomo is moving towards the event venue, it projects navigation images onto the floor for the people following behind Cocomo to guide them. Additionally, Cocomo obtained information

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about restaurant congestion via network and displayed information about available restaurants.

Cocomo can detect people located in the forward direction using the installed 2D LiDAR. Cocomo converts the scan results from LiDAR into a 2D image. A moving area that does not exist in the pre-created 2D map used for self-localization is identified as a person. When Cocomo detects a person within 3(m) in front, it stops moving and changes direction to ensure that the image projected onto the floor is facing the person as much as possible while tracking them. If multiple people are detected, Cocomo changes its direction while tracking the first person detected. As a result, the person being tracked becomes aware that Cocomo is watching them, prompting them to pay attention to Cocomo itself or to the images projected by Cocomo onto the floor.

While Cocomo is tracking a person and changing direction, it projects a stereoscopic image onto the floor. The projection of stereoscopic images is implemented by extending the method developed by the authors in reference [Ahn and Mizuno 2022]. In this technique, the principle of anamorphosis, which is one of the tricks of art, is used to achieve the projection of stereoscopic images. In this method, the positional relationship between the image and the observer's viewing position is important. In conventional methods, stereoscopic images are projected onto the floor from a moving cart, and the observers of the images either ride on the cart or stand in specific locations. Therefore, the positional relationship between the image and the observer was known in advance. On the other hand, in this study, the image and the observer move independently, so the positional relationship between the image and the observer's viewpoint is constantly changing.

In this study, we employed a method of tracking the user's viewpoint position in three dimensions using a Kinect camera installed on a Cocomo to obtain the positional relationship between the projected image on the floor and the observer's viewpoint. When Cocomo detects a person, it adjusts the orientation of the projected image on the floor to face that person as directly as possible. In doing so, the Kinect camera captures the 3D skeleton of people around the projected image, identifying the individual who is most directly facing the image. Subsequently, by determining the observer's viewpoint in three dimensions based on the head coordinates of the individual facing the projected image on the floor, a 3DCG simulation of how 3D objects appear from that viewpoint is conducted. This process generates stereoscopic images based on anamorphosis principles, which are then projected onto the floor. Since this process is performed in real-time, the target user can continuously observe stereoscopic images on the floor that change direction while tracking themselves. Therefore, floor projections can capture the user's attention, enhancing the effectiveness of navigation guidance and advertising.

3 EXPERIMENT

We implemented the system for event guidance at an international exhibition center and conducted a demonstration experiment by actually using it within the event for three days. In the experiment, we set up a circular route of approximately 100(m) near the entrance of the event venue. Cocomo automatically traversed the route while projecting images onto the floor. Figure 3 shows Cocomo moving while projecting images at the event venue.

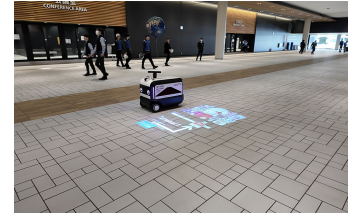


Figure 3: An experiment in an event.

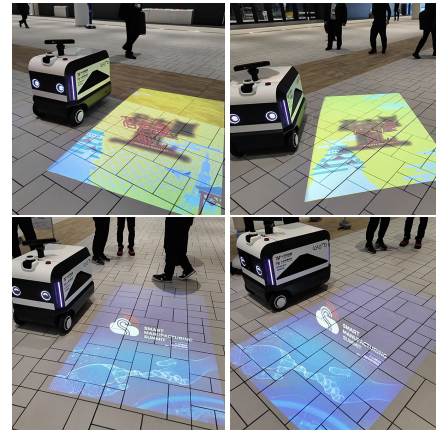


Figure 4: Stereoscopic projection mapping onto floor from Cocomo with user tracking.

When Cocomo detects a person within 3(m) in front of it, it automatically changes direction to track that person while projecting images. At that moment, we confirmed that the images projected onto the floor appeared as stereoscopic images to the subject by tracking the subject's viewpoint position. By employing the methodology of this study, stereoscopic projection mapping achieved a heightened sense of depth, as it combined the effects of anamorphosis with those of motion parallax (Figure 4).

The event drew over 10,000 attendees, and Cocomo, which projected images onto the floor, attracted a lot of attention from the attendees. More than 50 people actually experienced the system. The event organizers expressed their impression that the floor projections displayed by robots attracted a significantly larger audience compared to traditional floor posters for navigation, leading them to anticipate a higher effectiveness from this innovative approach.

Stereopsis achieved through anamorphosis and motion parallax may sometimes result in a diminished sense of depth due to discrepancies with binocular disparity. However, in footage captured by video, the influence of binocular disparity is eliminated, leading to a heightened sense of depth. Therefore, a significant number of people used their smartphones to capture Cocomo and the projected images. The potential for video dissemination through social media platforms could further enhance the advertising impact, resulting in a greater reach and effectiveness.

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